



Department of Computer Science

**BSCCS Final Year Project 2025-2026
Interim Report I**

25CS023

**Secure Peer-to-Peer Payment System on Android-Based Mobile
Devices via Vibration and Acceleration**

(Volume __ of __)

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Date : November 11, 2025

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Secure Peer-to-Peer Payment System on Android-Based Mobile Devices via Vibration and Acceleration

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ABSTRACT

The rapid growth of smartphone penetration in the modern century has laid the foundation for users to use mobile payment services for everyday transactions. The recent COVID-19 pandemic has significantly increased mobile payment usage and encouraged new account creation and mobile transactions in various peer-to-peer (P2P) mobile payment systems, such as Octopus and mobile wallets. The P2P mobile payment system allows two parties to facilitate secure payments by interacting in close physical proximity using mobile devices. The close physical proximity interaction is enabled by the Radio-frequency (RF) technology through Near Field Communication (NFC) and Bluetooth modules embedded in mobile phones. However, RF technology is vulnerable to several cyberattacks, such as eavesdropping and Man-in-the-Middle (MITM) attacks. It highlights the security problems of RF technology and encourages research on alternative solution for P2P communication. The project proposes a new radio-free P2P communication channel based on vibration for mobile payment systems. The built-in vibrator and accelerometer on the mobile phone are used for sending and receiving vibration data, ensuring that payments are conducted within a physical proximity distance. Moreover, to fulfill the security goals of the mobile payment system, such as confidentiality, integrity, and availability (CIA) measures, it is essential to integrate cryptographic frameworks for protecting the communication channel during transactions. However, vibration communication channels are low-bandwidth channels, making them unsuitable for traditional cryptography protocols. In this study, we propose a lightweight cryptography (LWC) framework that incorporates out-of-band key establishment using a passphrase and PBKDF2, and integrates the AES-CTR encryption scheme with HMAC in the communication channel. These LWC protocols aim to ensure the transaction is safe. Finally, functional, Performance, and Penetration testing of the system will be conducted to validate its viability and quantify its reliability for further studies on the vibration communication channel in the P2P mobile payment system.

Keywords—*P2P Payment System, Vibration-based Communication, Mobile System Security, Lightweight Cryptography*

1 Introduction

This section will begin by introducing the background and research gap. Moreover, it will also define the project objectives, justify the scope, and outline the deliverables for developing a secure Peer-to-Peer (P2P) payment system using a physical vibration channel. Lastly, it will provide an overview of the paper's organization.

1.1 Background of the Study

The increasing use of smartphones has fundamentally transformed the financial structure of modern society. Peer-to-Peer (P2P) payment systems have appeared as a keystone of modern digital payment transactions, enabling money transfer between users via mobile devices directly without using cash [1]. The recent COVID-19 pandemic has increased the acceptance of mobile payments, and the P2P payment market proportions reached \$1,889.16 billion in 2020 and are valued at approximately \$9,097.06 billion by 2030 globally [2]. Currently, Hong Kong has the highest percentage (46.7%) of interested users who use mobile payment system services monthly in the Asia-Pacific [3]. The P2P payment systems are believed to skyrocket and play an essential role in digital financial services worldwide.

Radio-frequency (RF) technology has seen rapid advancements in wireless network communication and mobile technology. Wireless payments using technologies such as Near Field Communication (NFC) and Bluetooth have become popular and are a standard protocol for short-range wireless communication channels. NFC has been adopted in P2P payment systems like Apple Pay, PayPal and Octopus [4, 5]. However, the RF-based approach requires particular hardware and can be vulnerable to compromise, such as eavesdropping and relay attacks [6]. Therefore, RF's drawbacks have encouraged the use of different channels for P2P communication, such as the High-Frequency Acoustic Wave channel [7] and the Visible Light channel [8]. Mobile phone device vibration and accelerometer sensing have also been proposed as a communication channel [9, 10]. These approaches may enable an offline physical P2P payment system to function without radio frequency.

1.2 Problem Statement

Previous research has successfully demonstrated the feasibility of using a smartphone's vibrator and accelerometer to establish data transfer through a physical channel [9, 10]. However, these studies have primarily focused on basic P2P communication for experimental purposes. Implementing the physical vibration channel, particularly in financial applications such as a P2P

payment system, remains largely unexplored. Few studies have examined the feasibility of the offline physical vibration-based P2P payment system. Proposing an innovative method for future research is necessary.

Moreover, deploying a security solution like an encryption scheme is crucial for protecting the P2P payment system against threats. However, the use of high-overhead traditional cryptographic protocols is not suitable in the case of vibration-based communication with severe constraints. Scholars have found that asymmetric encryption and public-key infrastructure (PKI) are particularly unsuitable. Their computational complexity drains resources, and a significant communication overhead is difficult for a low-bit-rate channel [11]. This creates a critical research gap in designing a lightweight cryptographic mechanism for high-latency, low-throughput vibration-based communication inside the P2P payment system.

1.3 Aim and Objectives

This project aims to tackle the insufficiency of using the vibration channel in the offline P2P payment system and design a lightweight encryption and key exchange mechanism for this communication channel. The objectives are as follows:

- To design a Vibration-Based Communication Protocol for the system: Develop an application-based algorithm for reliably encoding, transmitting, and decoding the data between two Android smartphones using the devices' vibrator and accelerometer.
- To develop a Lightweight Cryptographic Framework: Implement a simple security mechanism for the vibration channel. These include a secure custom method for secret key exchange and a simple encryption scheme for the transaction data.
- To evaluate the system performance and security: Test the final prototype of the application, and assess its performance, such as data rate, bit error rate, or latency; Analyze its security against a described threat model and identify possible vulnerabilities and limitations.

1.4 Project Scope and Deliverables

This project will develop an application on Android devices that will only use the built-in vibrator and accelerometer for communication, excluding other technologies like Wi-Fi or Bluetooth. The

project outcome will be a functional P2P payment System with the proposed communication protocol and cryptographic framework. The scope does not include development for other operating systems (OS) and defense against advanced physical attacks.

1.5 Overview of the Interim Report I

The rest of the paper is organized as follows. Section 2 will conduct a literature review to indicate the necessity of the work. Section 3 will present the preliminary design of the vibration-based P2P payment system. The final section concludes the plan and schedule for implementing and testing the system.

2 Literature Review

This section begins with an overview of existing P2P mobile payment system solutions, including NFC and Bluetooth. Next, a review of current vibration-based communication channels will be conducted, assessing their application and inadequacies for this project. Additionally, it will provide an evaluation of a cryptographic framework for vibration channels, followed by a critical examination of its limitations in the system.

2.1 Existing P2P Payment Technologies and Security Challenges

2.1.1 Near Field Communication (NFC) & Security Vulnerabilities

Nowadays, most P2P payment systems adopt the use of Radio-Frequency (RF)-based technology like NFC in personal electronic wallets (e-wallets) for contactless payment [13, 14]. NFC is a short-range P2P communication tool with radio frequency that combines Radio Frequency Identification (RFID) benefits with wireless communication technology like Wi-Fi. NFC is being introduced as a simple operation technology, as users only need to wave the mobile device close to another device to receive NFC signals, and the transaction can be completed immediately without other actions required [15]. This has fostered convenience and flexibility by tapping mobile devices for P2P payment without typing transaction details [16]. However, the protocol introduced a susceptibility to specific vulnerabilities, such as eavesdropping and Man-In-The-Middle Attack (MITM Attack), due to dependency on continuous internet connectivity [6, 17].

Regarding eavesdropping, NFC faces significant security challenges that create opportunities for alternative approaches. Previous research has shown that data on the NFC-enabled device can be intercepted at a distance by an attacker who listens to the communication via RF waves that transmit personal information or payment details and extracts the data from received RF signals during a P2P payment transaction. [18, 19]. This issue highlights security and privacy risks. Data branching can lead to unauthorized access, allowing attackers to gain unpermitted access to users' financial data throughout the NFC process. This violates the security principle of availability of "unauthorized denial of use" as the attackers can use and access the information without permission [20, 21]. Moreover, the data exposure risk has caused the attackers to collect and track users' information, and potentially utilize it wrongly to commit a crime or create a duplicate card that contains credit card data for card skimming [22].

Concerning the MITM attack, it could occur during NFC payments. Some researchers found that some NFC devices are connected to the cloud (Internet) channel instead of the traditional NFC,

where cryptographic operation happens inside the tamper-resistant hardware (Controller) to prevent cyber attacks [23]. They discover that if the communication channel between the mobile device and the cloud is not adequately secured, such as weak HTTPS protocols or unverified certificates, the attackers may intercept the traffic by creating untrusted CA certificates that fool users into signing them. and keep the MITM proxy for accessing the information during the transaction [23, 24].

The security loopholes of NFC highlight the need for an offline-capable payment solution that can operate without continuous internet dependency and radio-free communication protocols to avoid possible attacks.

2.1.2 Bluetooth & Security Vulnerabilities

Apart from NFC, Bluetooth is also a radio frequency (RF)-based technology that interconnects mobile devices with each other via P2P communication. Currently, the latest standard available is versions 5 and 6 with Bluetooth Low Energy (BLE), which helps to reduce the power consumption cost while maintaining short-distance communication [25]. Bluetooth operates on radio frequencies in the 2.45 GHz band, with a transfer rate from 1Mbps to 2 Mbps, and a transmission range from 10m to around 240m in omnidirectional format [25, 26]. Some researchers have implemented a “P2P-Paid system” that successfully conducts P2P mobile payment transactions on mobile phones via a nearby Bluetooth wireless network, showing the possibility of providing a functional P2P payment system using Bluetooth [4]. However, similar to NFC, Bluetooth relies on an RF signal that causes Confidentiality, Integrity, and Availability (CIA) security problems [27].

Regarding the Bluetooth security issues, previous research found that Bluetooth-enabled attacks are prone to security problems. For example, a Bluetooth sniffer can intercept the raw data being exchanged between two mobile devices [28], which could cause a serious problem with the confidentiality of transaction information since third parties may collect the sensitive data without notice. Furthermore, *Figure 2.1* shows that a MAC Spoofing attack can be made before the Bluetooth pairing between mobile devices. The attackers can clone a MAC address of the original device and impersonate it to manipulate data while in transition, showing that the information is being altered to confuse the receiver, violating the integrity [27, 29].

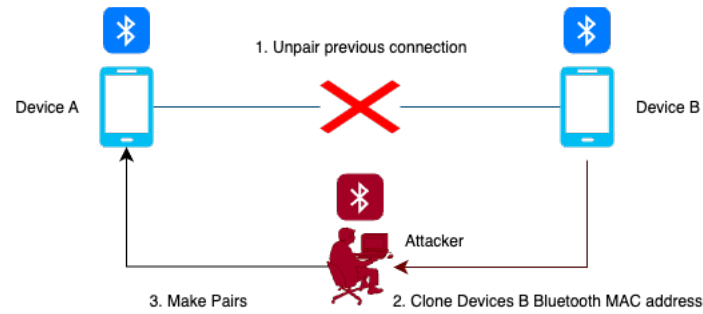


Figure 2.1: A MAC spoofing attack scenario using Bluetooth

Additionally, Bluetooth transmission must transfer in the form of RF waves. An attacker introduces sufficient RF noise into the Bluetooth network. Unwilling echoes can overwhelm the connection and prevent the establishment of communication [28]. This represents a Denial of Service (DoS) attack that blocks users from accessing a service by making it unavailable to users, which violates the availability principle in security [29].

The security inadequacies of Bluetooth emphasize the ongoing challenges in wireless P2P payment technology. These limitations create opportunities for a more secure and offline option, a P2P communication protocol in a payment system, like vibration-based solutions.

2.1.3 Secure Offline Covert P2P Communication technologies

To overcome the limitations of NFC and Bluetooth, research on alternative solutions such as physical covert P2P communication channels has demonstrated that it offer better physical security. The radio-free method needs close physical contact between mobile devices making it resistant to wireless attacks, such as eavesdropping [10]. Additionally, the lack of address discovery via the network or devices, as well as the short communication range, further safeguards data transfer [30]. Currently, there are some covert channels, for example Visual Light Communication (VLC) with Light Emitting Diodes (LED) approach proposed by Schmid et al. [8], where the approach sends information by a standard LED driven by a designed modulator that changes light intensity at a data rate to send out messages, and the receiver operates the LED in a photodetection mode that converts detected light into a photocurrent and decode it to a message. Additionally, Li et al. have proposed a communication channel based on high-frequency acoustic waves [7]. In which the sender plays high-frequency sound wave data through the air medium, then the receiver captures the audio and decodes the data. Moreover, many scholars have proposed the vibration communication channels [9, 10, 31, 33, 34, 35]. The data transmission sends a vibration pattern that encodes data using the smartphone's built-in vibrator, which is then decoded

by receiver using the smartphone's accelerometer. In the context of the P2P mobile payment system, vibration will be a more suitable methodology compared to other covert channels. Compared to the LED approaches, the LED requires additional hardware modules such as an Arduino board and LEDs embedded in the mobile device, which can be a burden for carrying and not user-friendly for users, and may also highly interfere with environmental factors (Such as sunlight or indoor lighting) during transactions [8]. Compared with the acoustic approach, although acoustic signals can be implemented through built-in smartphones' microphones and speakers, from an interference perspective, vibrations are less vulnerable to ambient noise, as busy or even quiet acoustic environments may degrade the sound channel [7]. The vibration on a table is relatively isolated from noise and light showing the channel is separated from the background environment, and it is a user-observable channel with strong physical proximity binding that helps to reduce the risk of possible conversion attacks[9, 10].

Therefore, to develop reliability and secure payment without connectivity requirements, this project proposed a sustainable vibration-based P2P transaction solution that ensures financial inclusion for those concerned about payment privacy and security.

2.2 Vibration-Based P2P Communication Technology

2.2.1 Vibration and Acceleration Data Transfer in Mobile Devices

Vibration-based P2P Communication is a non-radio-based, low-rate, and short-range communication that enables users to transfer data among smartphones by just putting one smartphone on another, similar to NFC communication [10, 31]. Communication requires the raw sensor data of the accelerometers and vibrators for the information sender and receiver, and this is feasible since this type of motion sensor is provided by the Android Platform [32]. The data transmission uses a built-in vibrator inside the smartphone to send a vibration pattern that encodes data decoded by the smartphone's accelerometer as a receiver. Current Vibration Communication methodologies adopt a similar decode mechanism using simple binary and different adaptive thresholding; however, they differ in the encoding phase [33]. There are three primary encoding methodologies: On-Off Keying (OOK)-based protocol [10], Amplitude modulation (AM)-based protocol [34], and Frequency Modulation (FM)-based protocol [35]. Previous research shows that OOK is considered more practical when applied to mobile devices, as it is simple to implement in all mobile phones by just switching the vibrator on/off to represent 1s and 0s using basic motors and API without considering the vibrator strength or frequency that may require additional

hardware or OS support [33]. Therefore, OOK is the most suitable method for mobile devices, considering the security and convenience of development.

2.2.2 Applications and Limitations in Previous Approaches

In the context of P2P communication, several research projects use the mobile phone vibrator and accelerometer for data transmission. Hwang et al. proposed VibComm, an offline Android application that uses vibration-based communication to exchange data between multiple users [9]. Yonezawa et al. have also developed Vinteraction, an information-transmitting application that leverages a combination of an accelerometer and vibrator using smartphones [31]. Both previous studies used vibration for P2P communication with OOK encoding and a similar adaptive thresholding decode mechanism, and it is offline and radio-free. Results found that 10cm is the minimum distance between two mobile phones, showing that data was successfully sent to other devices. However, they found that the transmission rate is low, limited to 0.09 packets or 10 bits per second, since a higher transmission rate will affect decoding accuracy [9, 31].

Roy and Choudhury have further proposed Ripple I and II [34, 35]. They have tried implementing AM and FM transmission protocols using a customized vibrator and accelerometer. Results show that they have pushed that transmission rate to 30 - 200 bits/s, and it is protected against the eavesdropper [34, 35]. However, it requires higher-level and self-developed mobile phone modules, which are impossible for ordinary users to develop. Finally, Zhao et al. have proposed LeaD [33]. They have modified the OOK encoding scheme to a bit group and decoded the bits using machine learning (ML) models for classification. The best ML model can boost the transmission rate to 40 bits/s with high accuracy [33].

The above scholars have inspired this project to adopt a Vibration-based scheme for P2P communication. However, previous studies have primarily focused on establishing a basic communication connection and transmitting simple strings only for experimental purposes only. They often assume that the channel itself is a trusted and safe one. Implementing a secure protocol for the physical vibration channel remains largely unexplored, particularly in financial applications such as a P2P payment system. The project also hopes to bridge this gap and show the possibility and feasibility of the security vibration-based P2P payment system.

2.3 Cryptography Framework for Vibration Channels

2.3.1 Security for P2P Mobile Payment Systems

P2P Mobile payment system security is essential for the users and service providers. The transaction data must be protected when it is used. The desired security goals in a mobile payment system include integrity, confidentiality, availability, authentication, access control, and non-repudiation [36].

To illustrate more, confidentiality requires that the data be restricted only to the parties involved in the transaction process and protection for passive attacks [37, 38]. Integrity ensures the system's accuracy during data processing, which is not being modified in a complete and timely manner [37, 38]. Moreover, non-repudiation ensures that it is a specific user's transfer transaction messages and prevents them from denying the services [38, 39]. Availability ensures the mobile payment system is accessible upon the user's request [38, 39]. Next, authentication requires verifying the users and the data origins [38, 39]. Finally, access control ensures only authorized persons can access the mobile payment system and blocks unauthorized personnel from accessing the payment system [38, 39]. This highlights the significance of a secure P2P mobile payment and the cryptographic framework, such as Symmetric and Asymmetric encryption, PKI, and digital signatures. These provide foundational tools to safeguard the mobile payment system and its security goals.

However, cryptographic challenges will be found in the low-bandwidth environment due to different constraints, such as a vibration channel. The sub-section below will discuss the relevant issues.

2.3.2 Traditional Cryptographic Challenges in Low-Bandwidth Environments

As mentioned in section 2.2.2, the vibration-based P2P communication only has a 10-200 bits/s transmission rate, which is significantly slower compared to other P2P communications used in mobile payment systems like Bluetooth with 1-2Mbps (Mentioned in 2.2.1) and NFC with up to 424Kbps [40]. Moreover, Mobile devices often work under constraints like limited processing power, memory, and battery capacity [41]. The twofold challenge has created a Low-Bandwidth and resource environment for the P2P payment system using vibration.

However, using high-overhead traditional cryptographic protocols is unsuitable in the case of a vibration P2P payment system, as it is difficult to process both vibration communication and cryptography in parallel. For example, previous research indicated that using asymmetric encryption and PKI in vibration-based channel devices leads to resource drainage due to computational complexity and significant communication overhead challenges for low-bit-rate channels [11]. Moreover, running a conventional cryptographic algorithm on a mobile platform consumes many resources, like CPU time and memory, making battery power drain fast and transactions slower [42, 43].

Yet, cryptography is necessary for establishing a secure mobile payment system. It must not be omitted to achieve the security objectives. Therefore, Lightweight Cryptographic (LWC) algorithms are essential in mobile payment systems to balance security and performance. The LWC framework provides strong security properties while minimizing the computational power to ensure a user experience and transaction time.

2.3.3 Effective Lightweight Cryptographic (LWC) Framework

The LWC Framework is not a new concept in cryptography. DES Lightweight extension (DESL) was first introduced in 2007 as a lightweight version of DES encryption, and the idea is simple: take DES and lighten it so that mobile devices can perform encryption at a low cost [44]. All the LWC shares common goals to run light and perform security goals, no matter how the algorithm is made. As Johnny pointed out, the LWC is feasible on mobile devices, and many lightweight API cryptography packages are supported for the mobile application environment [45]. At the same time, many LWC frameworks are supported. Our studies will focus on reviewing the algorithm used in our project.

Previous research has shown that symmetric cryptography is one of the lightweight solutions for the vibration communication channel [11]. Symmetric algorithms are generally faster and require less computational power than asymmetric ones, making them ideal for devices and communication channels with limited processing power, like smartphone motion sensors [46]. Advanced Encryption Standard (AES) is a standard symmetric LWC that provides security from a confidentiality and authentication perspective. AES is a block cipher that uses a fixed-length key and block sizes for encryption and decryption, and offers strong security guarantees with key sizes of 128 to 256 bits. Moreover, AES is part of many international and regional standards, which makes it a default option for secure communication in constrained environments [46]. From a

lightweight perspective, many smartphones currently include an AES hardware module that allows encryption to be executed without burdening the CPU and memory. In symmetric cryptography, the increase in key sizes does not significantly increase energy consumption, as the key expansion step in AES is only a small part of the total computation [46, 47]. The results concluded that AES may be suitable for the vibration communication channel, such as the LWC.

However, we have to consider the constraints of the AES approach to encryption when implementing it in the studies. First, the block size of the data will be a challenge. As mentioned in the previous literature, the key size is fixed in relation to the block size; if we choose a larger key size for better security, the block size will also increase. However, 128-bit blocks can be excessive for devices that only transmit small messages [46]. We must consider the tradeoff between security, computation power, and time when designing an AES encryption scheme for the vibration channel. The possible solution lies not in changing AES, but in changing the operation mode, as the operation's use will affect smaller or larger messages in a single block. In traditional protocols like AES-Cipher Block Chaining (CBC), when the message size is smaller than the fixed block size, padding is triggered and extra data is added to fill the blocks [48]. The method is not ideal for a vibration communication channel, as the limited bandwidth is wasted to process the redundant data created by padding. One of the modes that is more suitable in a lightweight mindset is AES-CTR (AES-Counter). The approach converts the block cipher to a stream cipher, which utilizes a counter (or nonce) with a key to construct a key stream and XOR it with the plaintext to form the ciphertext for communication (*Figure 2.2* below shows a typical flow of CTR mode) [49, 50].

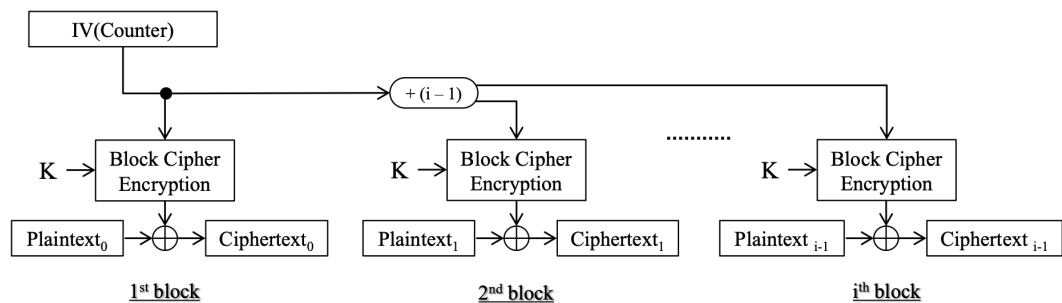


Figure 2.2: CTR Mode Encryption [50]

The CTR encryption shows that it is more parallelizable and doesn't require padding of the block while creating a ciphertext the same size as the small message being sent. It offers higher flexibility in determining the message size, which may suit a lightweight approach.

Moreover, Prior studies indicate that symmetric cryptography, which relies on exchanging a shared secret key between two devices, presents challenges in the vibration channel [11]. *Figure 2.3* shows that symmetric key cryptography uses the same private key for encryption and decryption, and both users require a pre-shared secret key to be established between parties before they can communicate securely [51].

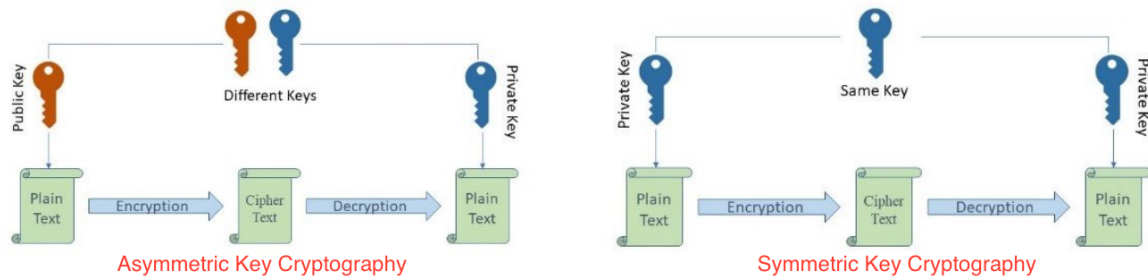


Figure 2.3: Asymmetric & Symmetric Key Cryptography processes comparison [51]

This presented the challenge of how to set up and exchange the key before communication in a lightweight method, as the energy consumption and resource demands of an AES depend not only on the process but also on the key setup cost [52].

A form of data transmission protection is password-based encryption, which involves using a password or passphrase input by the end users, and it serves as a cryptographic key for encryption [53]. It is an ideal lightweight methodology for key establishment since the protocol is designed for a few message exchanges and uses simple cryptography operations, which makes it suitable for a low-bandwidth channel [54]. Yet, it is very vulnerable from a security perspective since users will prefer weak passwords that are easy to remember; these passwords are generally short and easy to guess, like using their birthday as a password, which is highly susceptible to dictionary attacks that attackers try all possible passwords, until they find the correct one from a defined file [55, 56]. To reduce the weakness of the password, the “Password-Based Key Derivation Function version 2” (PBKDF2) has been introduced by the researcher to secure the password before use to establish a robust secret key. The PBKDF2 is denoted by:

$$DK = PBKDF2(p, s, c, dkLen)$$

The algorithm will finally output a derived key DK where p is the passphrase input by the user, a random salt s , an iterated loop counter c , and the secret key length $dkLen$ [57]. The PBKDF2 added salt to prevent rainbow table attacks and applied hashing like HMAC-SHA256 many times

on a defined iteration loop that establishes the secret key for the AES-secured. It is also a suitable approach in a lightweight context since it is simple to implement. The key length and iteration count can be tuned based on mobile device capabilities, which provide considerable flexibility for customization [58, 59]. Password-based encryption with PBKDF2 is a suitable lightweight method for key setup and exchange in our project.

Lastly, we should consider a cryptographic protocol to protect authentication and integrity during message transfer. Message authentication codes (MAC) are a symmetric cryptographic mechanism that allows two users with the same shared key to authenticate the data transmitted. *Figure 2.4* illustrates the whole simple process of the MAC protocol. First, the sender computes a MAC tag using the message and key K ; then, it sends the message with the tag as a block to the destination receiver. Next, the receiver computes a new MAC tag with the message and K and compares it with the MAC tag received. If it is equal, the receiver can conclude it is not altered during transmission [60].

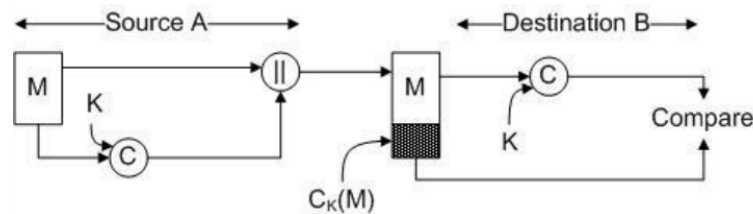


Figure 2.4: Overview of the Message Authentication Code [60]

From a security perspective, if we assume only the two parties know the secret key and the calculated MAC tag matches the received MAC tag, then:

- Integrity: The tag helps to verify that the message has not been modified since any alteration would cause verification to fail [60, 61].
- Authentication: The tag assures the message is from an authentic sender, proving that the data originated from a user who knows the secret key [60, 61].

The studies have proven that we should consider the MAC design when implementing a cryptographic framework. Nevertheless, given the limited processing power of the communication channel of vibration. We should also consider using a lightweight approach to the MAC. Previous studies have proposed a lightweight MAC using a “Hash-based MAC” (HMAC) approach to ensure security goals while keeping computation costs low in low-bandwidth channels, such as

those found in IoT devices [62]. The simple structure of the HMAC is relatively clean, where the function is $t = H(K, M)$, where t is the MAC tag of the Message M and the key K , and a hash function H using a compression algorithm like MD5, SHA-1, or SHA-256, and the complexity of the whole function depends on how the key and message are constructed [63]. Consider the design of lightweight HMACs by previous scholars, which only use hashing, the XOR operation, and concatenation of data. These are computationally inexpensive and reduce the communication overhead by minimizing the bits needed to transfer, but maintain the original security measures [62, 64].

To conclude, Section 2.3.3 provides a clear insight into LWC components to secure communication over constrained channels like vibration. Previous research supports the use of AES, out-of-band password-based encryption with PBKDF2, and the HMAC with a lightweight framework can help to fulfill the security goals of a P2P payment system. However, while these protocols are well-researched individually, there is a significant gap in integrating protocols within the vibration channel P2P payment mobile system. The project hopes to design and evaluate a lightweight cryptography framework for the secure P2P payment system.

3 Proposed Design

This section will provide a preliminary overall design of the vibration-based communication within the P2P payment system, as well as the design and integration of the cryptographic protocol into the communication channel. Moreover, it will also discuss how the system will be evaluated generally, including the testing procedures and setup.

3.1 P2P Vibration Mobile Payment System Overall

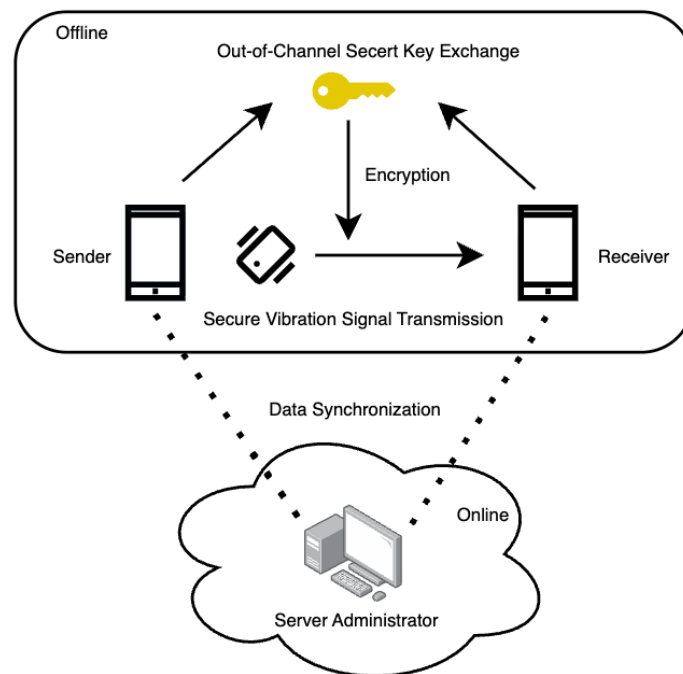


Figure 3.1: System diagram of the proposed solution

Figure 3.1 illustrates the overall system architecture of the vibration-based P2P payment system. The system will consist of three major components. The first components focus on the functionalities of the mobile payment system, including system features for users and the design of vibration communication. The second component concentrates on a payment management system for administrators, including transaction log recording and user data synchronization. The last component focuses on the lightweight cryptographic design, incorporating the secret key exchange and the encryption scheme for the vibration communication. All transactions can be completed in a physical and offline layer, except when data synchronization and transaction log saving are required from the server side.

We discuss the technical framework and libraries used for this project below:

For the coding environment, we will be using Android Studio IDE and the Kotlin programming language to develop each component in the project.

For the UI and functionalities of the system, the project will utilize Jetpack Compose, an open-source toolkit for building native UI on Android.

For the vibration communication channel, the Android Haptic APIs will be used to generate and control the vibration channel, and the Android Sensor Framework (*TYPE_ACCELEROMETER* & *SensorEventListener*) will be utilized to receive and listen to the vibration data.

For the Lightweight Cryptographic Protocol, the project will apply javax.crypto API for cryptographic operations such as SecretKeyFactory, Cipher and Mac.

For the database that stores user data and transaction records, Firebase will be used, as it is a flexible platform that can serve as a backend for various applications, including authentication, storage, and database management.

3.2 Functionalities of the P2P Mobile Payment System

Users can log in, log out, and create an account for the payment system. Moreover, they can also update their status, including their displayed username and passwords. This provides users with an authentication and customization function.

Moreover, users can check their payment history on the history screen, which will be categorized as “Pay” and “Receive payment” records. All payment logs will be sorted by date and time in descending order by default for improved display. Users can also search for specific transaction records by using keywords related to the transaction details.

Furthermore, users can send and receive payments from other users. If users choose “Send Payment”, the vibration encoder in the system will encode the vibration pattern of the payment based on the customized OOK algorithm and send it to the receiver for payment. If users choose “Receive Payment”, the vibration decoder in the system will listen to the vibration pattern from the transmitter and decode the payment vibration pattern using the adaptive threshold algorithm, and finally confirm the payment. In section 3.3, we will introduce the design of the vibration communication channel.

Prior to encoding or decoding the vibration payment pattern, the system requires both the transmitter and receiver to input a verbally agreed-upon secret passphrase. The system then generates a Pre-Shared Secret (PSS) key by processing this passphrase, which is used to encrypt and decrypt the vibration payment data. In Section 3.4, this paper introduces the design of the lightweight cryptographic protocol.

Lastly, if users are online (with an internet connection), they can update their account status and transaction log on the online server. From the server administrator's management side, they can check the transaction log and user status on the server.

Figure 3.2 below illustrates a use case diagram that describes the system's high-level functions and identifies the interactions between the system functions and its users.

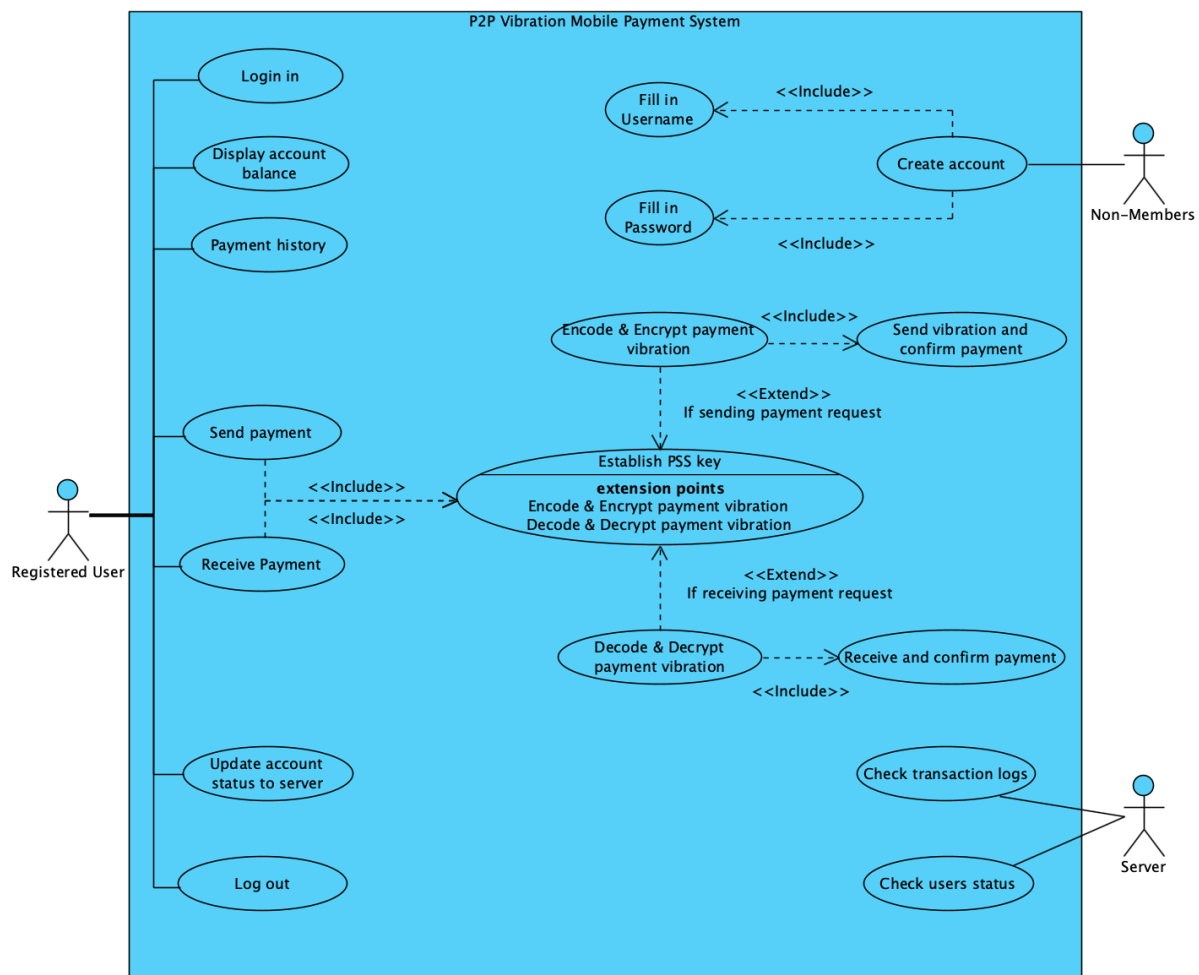


Figure 3.2: Use case diagram of the proposed solution

We should also consider a software engineering (SE) approach, such as design principles and patterns, before implementing the design of the vibration and cryptography.

First, we should consider the design principles. The Single Responsibility Principle (SRP) ensures that every class or object should only have one job, which helps to increase maintainability by making components highly reusable and easy to test. For example, vibration design should ensure that the vibration encoder is only responsible for encoding the payment request to a vibration pattern and the decoder should only perform the reconstruction of the original data of the vibration.

Furthermore, the Open-Closed Principle (OCP) should be considered; the function in the system should be open for extension and closed for modification. OCP helps minimize the impact on existing code when making changes, and makes it easy to add new functions. Abstraction and subclassing techniques can help to achieve the principle. For example, the Lightweight Cryptographic Protocol always needs to encrypt and decrypt the message. Their used attribute and behavior will always remain the same, and the only difference is the cryptographic algorithm used in each process. Therefore, the OCP can ensure that the class is always open for extending the use of the protocol, while maintaining the behavior and attributes defined in the design.

Lastly, this project should also consider the KISS principle, which emphasizes keeping the project simple and reliable to solve complex problems. For example, in this project, we are evaluating the secure vibration channel but not developing a complex mobile payment system, so we should implement a simple function to demonstrate each use case using the proposed solution.

Next, we should consider the design pattern. The singleton pattern should be considered; this pattern ensures that only one instance of a class exists in the entire application. It provides better control as there is only one vibration controller for physical vibration and prevents duplicated parts from performing the same tasks.

Additionally, the observer pattern is considered, where there is a listener is notified of changes from the publisher. In our project, the UI of each component should observe the update of the variable and update the display information automatically. For example, the home menu observes the changes in the user's balance and displays the actual balance of the user.

Lastly, to ensure a program performs tasks in the background without halting its main execution, it is necessary to prevent the vibration channel from requiring the user to idle the mobile phone for sending and receiving vibration patterns, as this would completely freeze the interaction between the application. Kotlin Coroutines are used for structured concurrency and thread confinement. For example, the send payment coroutine in HomeMenu runs in the background, but updates the transaction status concurrently to prevent memory crashes or loss.

Figure 3.3 illustrates a simple class diagram of the proposed design, which incorporates the considered design patterns and principles.

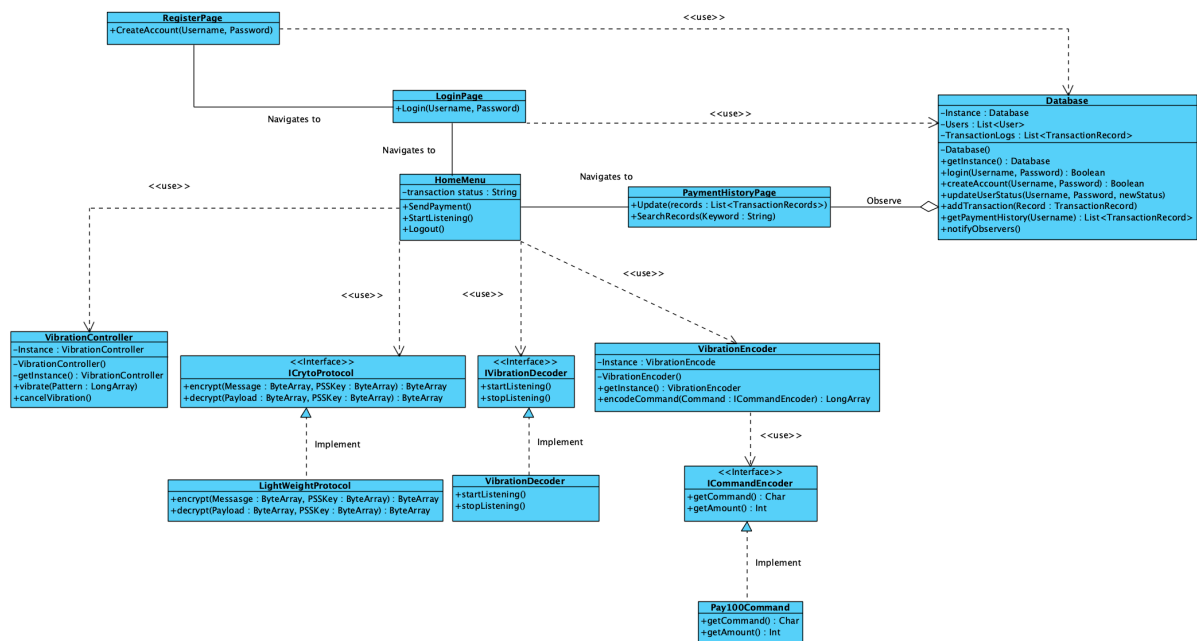


Figure 3.3: Class diagram of the proposed solution

3.3 Vibration Communication Channel Design

An overview of the design of vibration communication will be presented below. It is separated into two major components: 1) Vibration Encoder and 2) Vibration Decoder. Moreover, there are two minor components to support the encoder and decoder: the Vibration Controller and Accelerometer Monitor, which utilize the built-in vibrator and accelerometer inside the smartphones. Figure 3.4 below shows the overall design of the vibration communication channel.

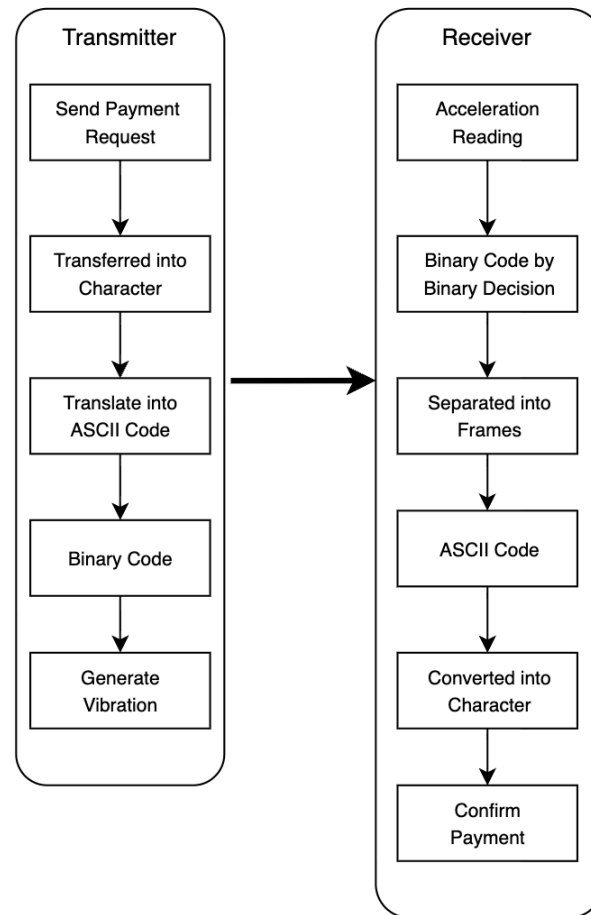


Figure 3.4: An overview of the vibration communication channel

The Vibration Controller is a simple and low-level hardware component that directly interacts with the mobile device's haptic motor. It is responsible for receiving a pre-defined vibration pattern from the encoder and executing it on the device vibrator. Moreover, it also ensures the vibration strength is set to maximum (255) only during vibration and 0 when it is paused.

The Accelerometer Monitor serves as the hardware interface for data collection and conversion. It detects the physical vibration by listening to the device's built-in accelerometer. It collects the raw sensor data and stores these readings in temporary buffers, then transfers the data to the decoder for processing. Below shows the details of the two key actions.

- **Signal conversion:** It collects the raw X, Y, Z -axis data from the accelerometer and converts it to a single-scale magnitude. The formula is defined as: $M = \sqrt{X^2 + Y^2 + Z^2}$.
- **Data buffering:** It helps to transfer all the object data to a list and pass it to the Vibration Decoder for analysis.

The Vibration Encoder is a primary component for the transmitter. The function of the encoder is to transfer a payment request into the vibration pattern. Below shows the designed process:

- **Data Framing:** The encoder first converts the payment request into characters, such as “a”, and then converts it to a 7-bit ASCII code. Then, we will construct the vibration pattern into a frame format of “Start (00010) + Data (1100001) + End (00011)”, resulting in a total of 17-bit data frame.
- **Modulation:** The data frame will then be modulated using the OOK scheme. Each bit is allocated into a 1-second time slot, and the data rate is fixed at 1 bit per second (bps). Where bit ‘0’ is a 1000ms pause (No vibration), and bit ‘1’ is a 200ms vibration, and padded with 400ms on either side (400ms pause + 200ms vibrate + 400ms pause) to fit the 1000ms window.

The results are a 17-second vibration sequence, such as [1000, 1000, 400, 200, 400, ...], which is then passed to the Vibration Controller for execution.

The Vibration Decoder is a primary component for the Receiver. It analyzes the raw sensor data collected by the Accelerometer Monitor to reconstruct the original data. This process is triggered automatically after a fixed buffering period and involves two main stages.

- **Bit Reconstruction:** A function is available for converting analog accelerometer data back into a digital bitstream. It iterates through the buffered list in a discrete 1-second window corresponding to the synchronous bit rate. For each window, it checks if the magnitude exceeds a set vibration threshold a sufficient number of times. If vibration is detected, a “1” is appended to the binary string and vice versa.
- **Pattern Matching:** The constructed binary string will be passed to another function that searches for the 17-bit data frame to verify its validity. This is achieved by finding the start marker and subsequently checking the 12 bits for the end marker. The data frame is then extracted for processing.

Finally, if only a single valid pattern is found, the decoder converts the ASCII value back and identifies it as the payment command to process the required transaction. If multiple valid patterns are detected due to signal noise, the user is presented with the list of possibilities to make the final selection of the pattern.

3.4 Lightweight Cryptographic Protocol Design

The vibration communication channel, as described in Section 3.3, provides a novel method for data transmission. However, the pure data frames are transmitted without any protection, which makes them vulnerable to several attacks and does not fulfill the security goals of the mobile payment system. This section provides the details of the LWC design to secure the channel.

It is essential to examine the security challenges that the channel may face and highlight the necessity of a cryptographic design, as well as a critical consideration of the design process.

First, it may be vulnerable to eavesdropping if the attacker is also in physical proximity and could intercept the vibrations, decode the messages, and discover the payment request from the users. Next, it may also be exposed to replay attacks, as the attacker can record a valid vibration pattern and replay it later to authorize a duplicated payment.

Although the above attacks are mostly impossible, as users can easily discover third parties and ensure the identity of the sender and receiver in a close physical environment, but we consider possible attacks from a theoretical perspective. To prevent potential attacks and fulfill the security goals outlined in the literature review (Section 2.3.1), we should implement a simple cryptographic algorithm to secure the communication channel.

Moreover, this project should also consider the framework used, as described in the literature review section (Section 2.3.2). Since the vibration channel is a low-bandwidth (1 bps) channel, we should consider using the lightweight cryptographic frameworks stated in section 2.3.3. Symmetric key cryptography should be used. We should also consider how and which protocol should be implemented in a lightweight method.

Based on the findings of the literature review (Section 2.3.3), the framework is constructed based on three standard lightweight symmetric protocols listed below:

- **Key Derivation:** To solve the key exchange problem of the symmetric encryption of AES, both the transmitter and receiver will first verbally agree on a shared passphrase. After exchanging the passphrase, the system will process it using PBKDF2 with a static salt and a fixed iteration count. The hard-coded method ensures that both devices generate the same 128-bit PSS key.

The PSS key will be stored only temporarily in the storage and will be removed after the transaction is complete, and it is not expected to be reused.

- **Encryption Scheme:** Considering the transmission time of the vibration channel, we will consider using AES-CTR. This converts the AES block cipher into a stream cipher, enabling it to generate a keystream of the exact length required to encrypt the data, thereby avoiding padding that could result in increased transaction time.
- **Integrity and authentication:** To prevent tampering, a HMAC-SHA256 is used. The 256-bit output is truncated to 16-bit tags and concatenated into the data.

Below, this project proposes a new lightweight cryptographic framework with the protocols reviewed. The notations used to describe the protocol are defined in *Table 3.1* below.

Notation	Description
Passphrase	An out-of-band, secret shared passphrase between the two users
PSS_Key	Pre-Shared Secret key
PBKDF2	Password-Based Key Derivation Function version 2
M	Original data frame of the payment message
P	Plaintext combines the message and the tag
C	Ciphertext that encrypted the plaintext
Payload	The final data stream transmitted via vibration
N	Nonce
T	The authentication Tag
K	The pseudorandom Keystream generated by AES-CTR
HMAC	Hash-based Message Authentication Code
truncate	A standard function that shortens a binary bitstring
AES_CTR	AES in Counter mode
$\oplus$	The bitwise XOR (Exclusive OR) operation
$\parallel$	A concatenation operation

Table 3.1: Notations

The original 17-bit data will not be transmitted at first. It is encrypted and wrapped into a new secure data. The new total data transmitted over the vibration channel is 41 bits, including the

security measures. *Figure 3.5* below shows the new data structure after encryption, which will be sent to the receiver and the details of the modified plaintext.

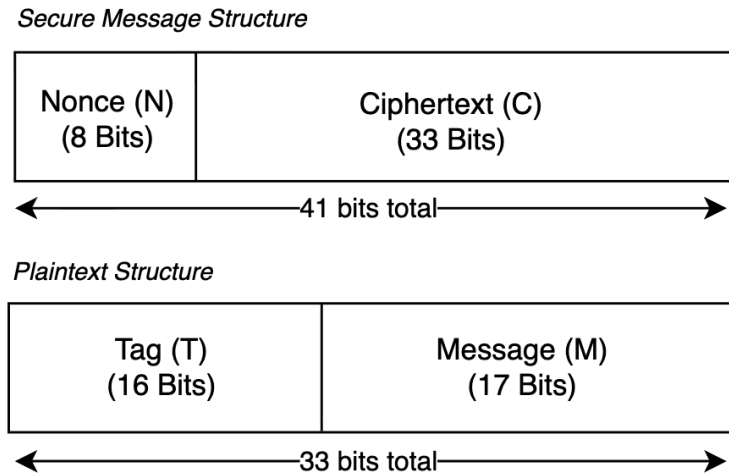


Figure 3.5: Final Data Stream Structure

An 8-bit nonce (N) is a random initialization vector included in the message to ensure that every transaction is unique and distinct. The system will generate a random number from 0 to 255 and get the 8-bit binary string as the nonce to fit the design of the OOK scheme for the vibration channel. Inside the 33-bit plaintext, it includes a 16-bit HMAC tag (T), which is truncated from HMAC-SHA256, and the original 17-bit vibration message (M).

A general workflow of the proposed schemes is detailed as follows:

First step: Both the sender and Receiver verbally agree on a passphrase, and the system will derive an identical 128-bit PSS Key for both users.

$$- PSS_Key = PBKDF2(passphrase)$$

Second step: The sender will first construct a 17-bit message for the payment request and generate a unique 8-bit nonce. Then, the system will generate a 16-bit HMAC integrity tag using the message, nonce, and the PSS key, and also generate a 33-bit keystream with AES-CTR using the PSS key and Nonce. Next, the system combines the message and the tag into plaintext and encrypts it using the keystream to produce the ciphertext by the XOR operation. Finally, the system assembles the nonce and ciphertext into the 41-bit payload and passes it to the vibration controller to send out.

$$- T = truncate(HMAC(PSS_Key, M || N), 16 \text{ bits})$$

$$- K = AES_CTR(PSS_Key, N)$$

$$- P = M || T$$

$$- C = P \oplus K$$

$$- \text{Payload} = N || C$$

Third step: It is just a reverse step of the second step, where the VibrationDecoder reconstructs the 41-bit stream from the vibrations and splits the received payload into an 8-bit nonce and a 33-bit ciphertext. Then the system uses the PSS key and the received nonce to generate the same keystream and decrypt the ciphertext back to the plaintext. Next, the system will split the 33-bit plaintext into the received 16-bit tag and the 17-bit messages, and calculate its own expected tag for verification. If the expected tag matches the received tag, then it indicates the authentication and integrity of the data, and the transaction will be processed. Otherwise, the transaction will be terminated by the protocol.

$$- K = \text{AES_CTR}(\text{PSS_Key}, N')$$

$$- P' = C' \oplus K$$

$$- T' = \text{truncate}(\text{HMAC}(\text{PSS_Key}, M' || N'), 16 \text{ bits})$$

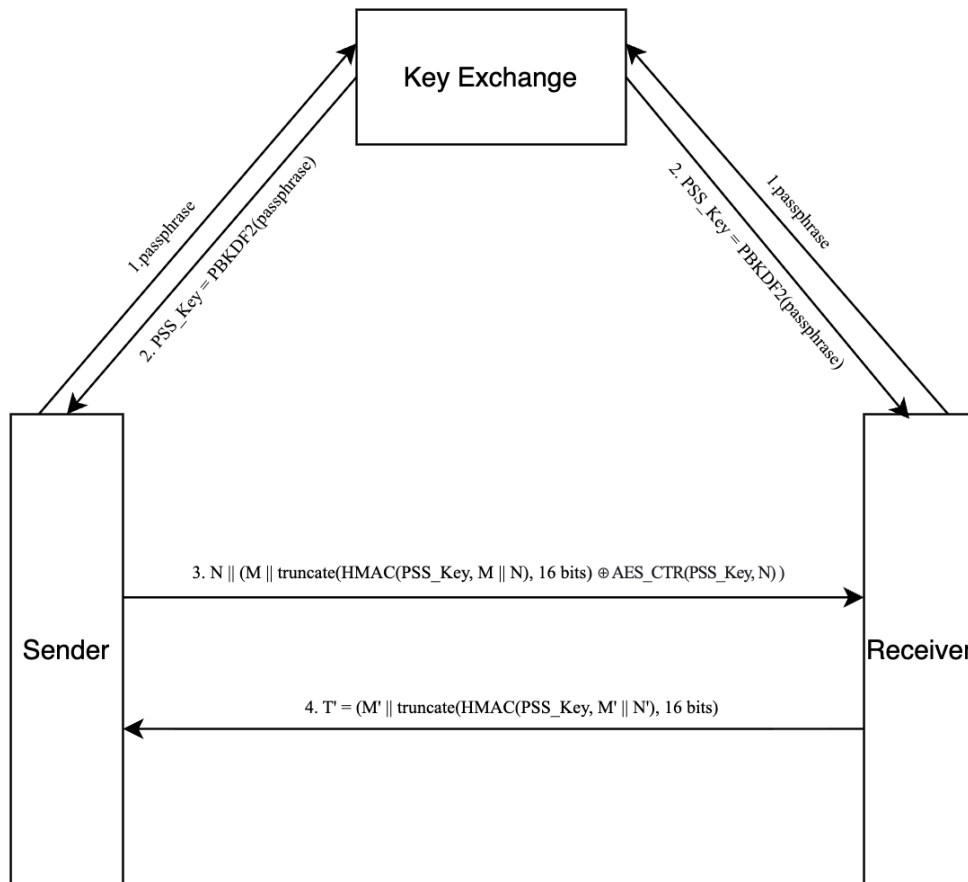


Figure 3.6: Description of the proposed protocol

3.5 Performance Evaluation Design

The Final system will be tested on two Android phones, the Samsung Galaxy A8+ and the Huawei P30 Pro, in a controlled environment, such as on a table at close distance (Within 10 cm). Below will show the details of the testing that will be conducted in the project. A Bottom-up approach will be used to test the system's functionality, followed by performance and penetration tests for testing the vibration algorithm and security protocols.

Unit Testing

During system development, we will perform unit tests for each developed individual class according to the functions specified in Section 3.2.

Integration Testing

After completing the development of each phase, we will integrate the subset of each class into a subsystem, like a combination of the UI and the vibration transaction. The purpose of it is to ensure that the functions in the subsystem interact normally.

System Testing

In the final phase of implementation, we will integrate all subsystems into a comprehensive P2P vibration-based mobile payment system to test its functionality without faults.

Below are some sample functional test cases that will be performed:

User Create Account	
Requirement: User type the username and a password longer than 6 character for register account	
Test Case:	Expected Result:
User input a valid username and valid password Username is not being registered	User created an account in local database and forward back to the login page for login
User input a valid username and valid password But the username is being registered	Alert ("Username is being registered, please use another name!") Stay in register page
User input a invalid username (Exceed String Limit or too short) and valid password	Alert ("Invalid username, Please try again!") Stay in register page
User input a valid username but invalid password that break the security rule	Alert ("Invalid password, Please try again!") Stay in register page

User login in account	
Requirement: User type the valid username and password to login in	
Test Case:	Expected Result:
User input a valid username and valid password (Account is existed)	User Logged in, forward user to Home menu display welcome message User data retrieved from local database
User input a invalid username and valid password	Alert ("Please input an valid username!") Stay in login page
User input a valid username and invalid password	Alert ("Invalid password, Please try again!") Stay in login page
User input invalid username or password multiple time	Alert ("Login in function locked, please try again later!") Back to mobile home page

User settings (For updating the username or password)	
Requirement: User need to update valid username or password for changing account status	
Test Case:	Expected Result:
User input a valid username or valid password and click "Update Status"	Alert("Update successfully!") Update user status in local database
User input a invalid username (Repeat username)	Alert ("Username is being register, please try again!") Stay in setting page
User input a invalid password that break security rule	Alert ("Invalid password, Please try again!") Stay in setting page

Checking payment history	
Requirement: User want to check transaction record and enter payment history page	
Test Case:	Expected Result:
User has payment record	Automatically display the transaction record in the page
User has no payment record	No payment record in the page

User Data Synchronization	
Requirement: User want to update the account status or transaction record to the server admin	
Test Case:	Expected Result:
User click "Upload Account status and transaction records" and have newest record that not yet update	Alert("Update successfully!") Update user status in online database
User click "Upload Account status and transaction records" and don't have newest record for update	Alert ("No new record required to update, please try again later!") Return to home menu

Server Management page	
Requirement: Administrator want to check the transaction records and users account status	
Test Case:	Expected Result:
Admin click “Check User account status”	Display all the user in the list
Admin click “Check transaction record” from one of the user in the list	Display selected user transaction record

PSS Key generation	
Requirement: User’s establish the share key using passphrase before performing transaction (Sender and Receiver)	
Test Case:	Expected Result:
Both user’s enter the exact same secret passphrase	Both devices process the passphrase and generate the identical PSS key for processing transaction later
One of the user enter the different secret passphrase	The generated PSS key will be different, transaction will fail alert ("Authentication Failed!" or "Corrupt Data!") during transaction process

Send Payment Transaction	
Requirement: Sender send payment transaction to receiver (Assume PSS key is successfully establish for both users and place phones on the same surface)	
Test Case:	Expected Result:
Tap “Send Payment”	The payment request will be encrypted and the vibration encoder will encode the data based on the OOK algorithm to vibration pattern. The phone will physically vibrates
User tap “Cancel payment request” after requesting payment	The transaction request will be terminated, no money will be paid

Receive Payment Transaction	
Requirement: Receiver receive payment transaction to sender (Assume PSS key is successfully establish for both users and place phones on the same surface)	
Test Case:	Expected Result:
Tap “Receive Payment” (HMAC Tag same)	The vibration decoder will listen and capture the vibration pattern and and decodes it using the adaptive threshold algorithm, decrypt the data, calculate the expected tag and confirms the payment
Tap “Receive Payment” (HMAC Tag not the same)	After calculatd the expected tag but it is not same with the received tag, transaction will fail and terminated, alert ("Authentication Failed!")
User tap “Cancel payment receive” after receive payment	The listen of vibration will be terminated, no money will be receive

Performance Testing

We will also test the package delivery rate and the accuracy of the transaction transmission using the designed vibration encode and decode algorithm. If time allows, we will also consider testing the algorithm under different communication distances and surface materials, not just in a controlled environment. It helps validate the design's viability and quantify its reliability for further studies on the vibration channel in the mobile payment system.

Penetration Testing

Penetration testing (Pen test) is a simulated cyber attack designed to exploit vulnerabilities in a system before it is publicly released. It helps to validate if the theoretical cryptographic design is implemented securely in practice and achieves the security goals.

The pen test will focus on the security of the vibration communication channel and the key establishment. Since the key is used to encrypt the vibration data in each transaction, it is expected that the key will not be reused. Hence, the ethical hacking attack is expected to mainly target the proposed security design to test whether it is vulnerable to attacks such as replay attacks and eavesdropping.

4 Plan and Schedule for Implementation and Testing

This section will provide information on how the system will be developed, including the software development approaches and stages for implementation and testing.

4.1 Software Development Approaches and Stages

The project follows an Agile software development approach, with iterative development cycles. Implementation has been distributed into five phases (1-5), allowing for greater flexibility. The key milestones and schedule are outlined below:

a) Phase 1: P2P Payment System Interface (September 2025)

- **Goals:** Develop a P2P payment system interface for two Android phones
- **Key Deliverables:** A functional payment interface with basic transaction operations
- **Improvement on the previous phase:** N/A
- **Testing:** Unit testing on UI interfaces
- **Documentation:** Project Plan

b) Phase 2: Basic Vibration channel development (October – November 2025)

- **Goals:** Set up the development environment and implement core vibration transmission and reception between two phones
- **Key Deliverables:** A basic Android app that can encode a payment command into a vibration frame and decode it back on the receiver device
- **Improvement on the previous phase:** Implementing vibration and payment request into the UI interface
- **Testing:** Unit and integration testing on the UI interfaces and the vibration communication channel
- **Documentation:** Interim Report I

c) Phase 3: Cryptographic framework development & Integration (December 2025 – January 2026)

- **Goals:** Design and develop the passphrase-based key exchange and AES encryption with the HMAC scheme
- **Key Deliverables:** An Android app prototype that can perform a secure key exchange and send an encrypted message via vibration

- **Improvement on the previous phase:** Integrating the proposed secure protocols into the vibration communication channel and UI interfaces
- **Testing:** Unit and integration testing on the UI interfaces and the vibration communication with secure cryptographic protocols
- **Documentation:** Interim Report II preparation

d) Phase 4: System Integration & Testing (February 2026)

- **Goals:** Integrate all components into a single, comprehensive system and conduct testing and evaluation of the system
- **Key Deliverables:** A fully functional end-to-end secure P2P vibration mobile payment application
- **Improvement on the previous phase:** Improve the readability and quality of the source code
- **Testing:** System testing on the whole system, and perform preliminary performance and Penetration testing
- **Documentation:** Interim Report II & Final Report preparation

e) Phase 5: Final Preparation & Documentation (March 2026)

- **Goals:** Prepare the final report, create a demo video, and fine-tune the source code for final submission. Moreover, preparation for the FYP presentation.
- **Key Deliverables:** Final project report, complete source code, demonstration video, and a presentation slide
- **Improvement on the previous phase:** Improve the readability and quality of the source code
- **Testing:** Final performance and penetration testing
- **Documentation:** Same as Key Deliverables

Below is the Gantt Chart for the Final Year project (Divided into documentation & System development Gantt Chart):

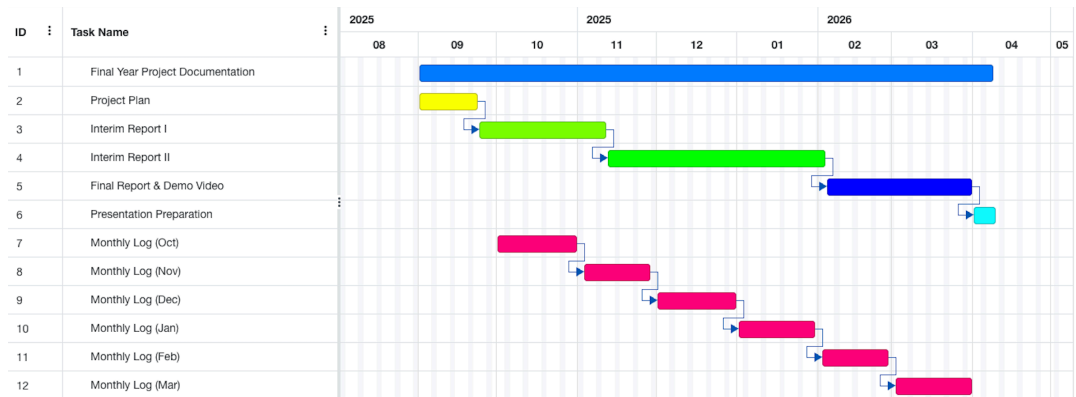


Fig.4.1: A Gantt Chart for the Documentation Part of the Project

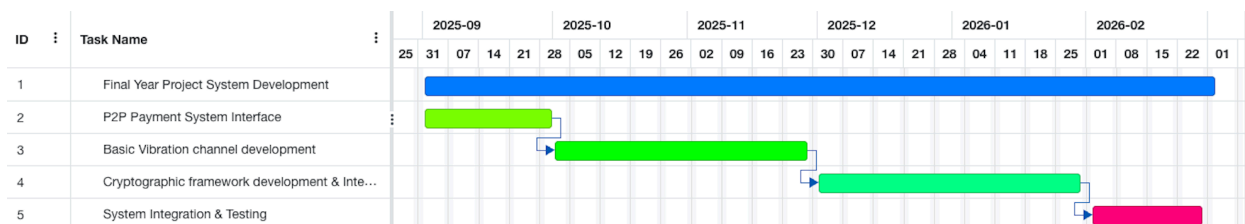


Fig.4.2: A Gantt Chart for the System Development part of the Project

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Appendices

Appendix A - Monthly Log

October 2025

I will provide my progress report in three sections each month.

It includes: 1) Documentation 2) Implementation 3) Expectations/ goals for the next month.

For Documentation, I have reviewed the comments on the project plan from my supervisor and have completed the draft of the introduction and literature review part for the interim report I this month. Moreover, I have brainstormed the design and testing methods for the overall system.

For implementation, I have completed the UI coding part of the P2P payment system. However, I'm kind of behind schedule due to the busy midterm weeks and assignments. For the vibration channel part, I have completed the encoding scheme to generate vibration. However, I am still stuck on a minor issue related to detecting vibration by calculating magnitude data for the decoding scheme.

Expectations to be complete next month:

- Complete the draft of the proposed design and implementation/testing schedule of the interim report 1 and submit it to my supervisor for suggestions and reviews
- Complete basic vibration communication channel code development (Without encryption)
- Study on proposed lightweight cryptographic protocols and consider how to integrate them into the communication channel

November 2025

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